Micro-Project Report: Computational Implementation of Real-Time Optimization Problems

24MAT382 - Computational Optimization and Applications

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1 Problem Statement

The objective of this micro-project is to validate the ability to identify a real-time problem, formulate it mathematically, and implement a computational solution using Python. Specifically, this project addresses the real-time scenario of designing an optimal emergency supply distribution network for Campus City Logistics. The network must minimize operational costs while satisfying all facility demands, demonstrating a direct application of Integer Programming concepts covered in the course modules. The model evaluates three potential warehouse locations to select exactly two, ensuring operational redundancy. These warehouses distribute essential supplies to six critical campus facilities while strictly adhering to a maximum annual budget constraint of $1,500,000, respecting individual warehouse storage capacities, and ensuring 100 percent of daily demands are completely fulfilled.

2 Mathematical Formulation

This problem is formulated as a Mixed-Integer Linear Programming (MILP) model. The modeling phase requires the correct identification of decision variables, an objective function, and constraints. The sets include critical facilities indexed by i and potential warehouses indexed by j. The parameters consist of the annual demand for each facility, the annual capacity of each warehouse, the annualized fixed costs, the transportation cost per unit, and the maximum budget limit. The decision variables are defined as a binary variable y_j, which equals 1 if a warehouse is selected and 0 otherwise, and a continuous variable x_{ij} representing the exact number of units shipped.

The objective function seeks to minimize the total annual cost, represented by the equation:

$$\text{Minimize } Z = \sum_j (FC_j \cdot y_j) + \sum_i \sum_j (C_{ij} \cdot x_{ij})$$

This is bound by four constraints. First, demand satisfaction ensuring every facility receives its required amount ($\sum_j x_{ij} = D_i$). Second, capacity limits ensuring shipments do not exceed warehouse storage ($\sum_i x_{ij} \leq K_j \cdot y_j$). Third, a redundancy requirement

forcing the selection of exactly two warehouses ($\sum_j y_j = 2$). Finally, a budget constraint capping total costs:

$$\sum_j (FC_j \cdot y_j) + \sum_i \sum_j (C_{ij} \cdot x_{ij}) \leq B$$

3 Methodology

The computational implementation utilizes Python to solve the optimization problem. Data preprocessing was handled using the Pandas library to ingest and clean the datasets containing facility locations, demands, and transportation costs. The core algorithm relies on the correct use of the PuLP library to construct the MILP model and manage the logical flow of the algorithm. PuLP effectively combines discrete binary variables for warehouse selection with continuous variables for routing flows, guaranteeing mathematical optimality through its underlying solvers. Additionally, the Folium library was integrated to plot geographic coordinates and dynamically visualize the optimal supply routes. The complete executable Python source code and corresponding datasets are hosted in an organized GitHub repository as per the submission deliverables.

4 Results and Analysis

The execution of the Python script yielded an optimal solution that was carefully interpreted and verified against all mathematical constraints. The algorithm mathematically rejected the East Gate warehouse due to unfavorable geographic positioning, instead selecting the North Campus and South Campus warehouses. This successfully fulfilled the redundancy requirement. The optimized network generated a total annual cost of \$959,466.05, operating safely below the \$1,500,000 budget cap and generating an operational surplus of \$540,533.95. The capacity constraints were fully respected, with the North warehouse achieving a 47.5 percent utilization rate and the South warehouse reaching 22.6 percent. The resulting transportation matrix prioritized low-cost routes, verifying that 100 percent of the demand for all six target facilities was met efficiently.

🚀 Initializing Optimization Engine...

✅ OPTIMIZATION SUCCESSFUL

💰 Total Annual Cost: $959,466.05
📦 Total Units Delivered: 98,185 units
📈 Avg. Cost Per Unit: $9.77 / unit
🏦 Remaining Budget: $540,533.95

🏢 Warehouse Utilization Report

Warehouse	Status	Fixed Cost	Utilization
WH_NORTH	🟢 OPEN	$322,000	47.5%
WH_SOUTH	🟢 OPEN	$283,500	22.6%
WH_EAST	🔴 CLOSED	$360,500	0.0%

🚚 Optimal Routing Schedule

From	To	Units Shipped	Transport Cost
WH_NORTH	MED_CENTER	29,565	$113,233.95
WH_NORTH	ENG_BUILDING	10,950	$38,653.50
WH_NORTH	DORM_A	17,885	$61,882.10
WH_NORTH	LIBRARY	10,950	$40,186.50
WH_SOUTH	SCIENCE_HALL	10,220	$35,974.40
WH_SOUTH	DORM_B	18,615	$64,035.60

Figure 1: Optimization Engine Output: Financial summary, warehouse utilization, and routing schedule.

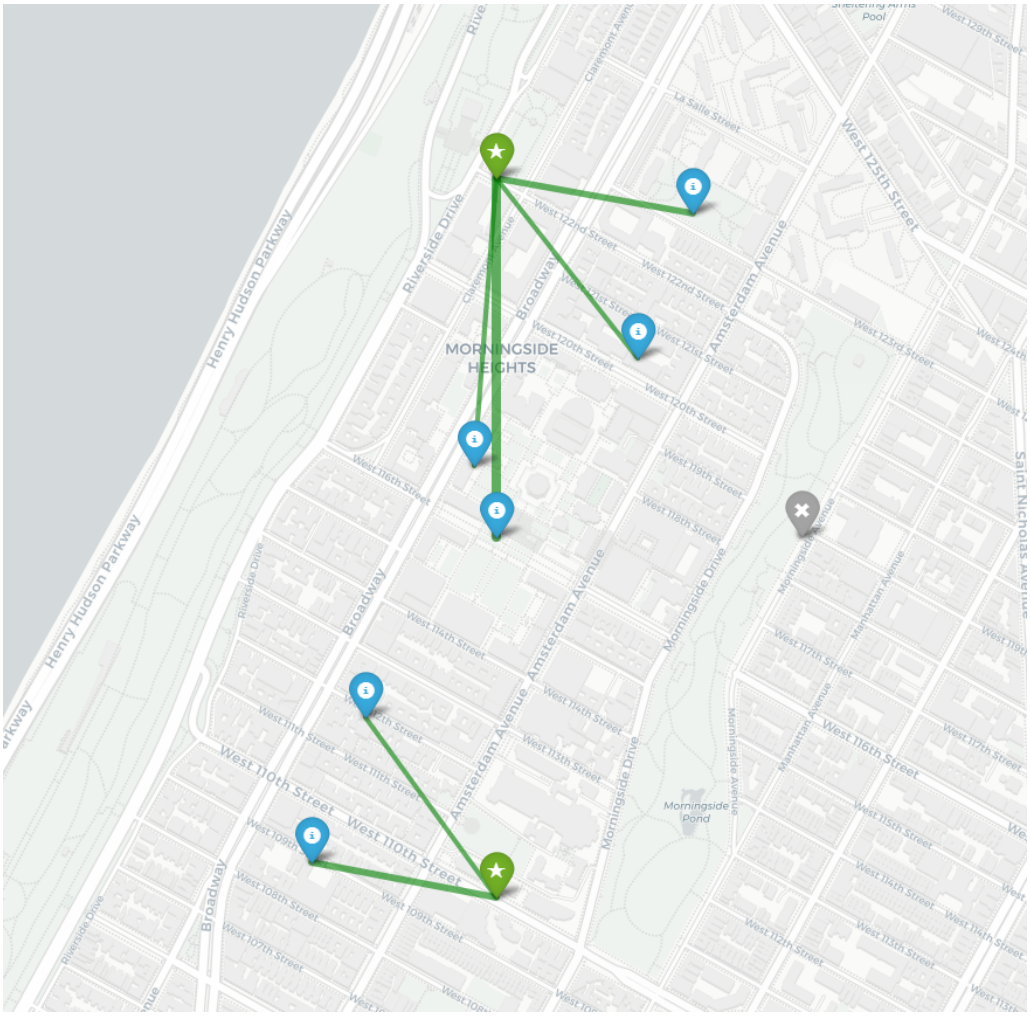

Figure 2: Interactive geospatial map displaying the selected North and South warehouses and optimal supply routes.

5 Conclusion

The implementation of this MILP optimization model successfully modernizes the Campus City logistics network. By applying a mathematically sound methodology to facility location and routing, the project demonstrates that operational overhead can be significantly reduced while improving service reliability. The model guarantees demand satisfaction, respects capacity limits, and maximizes budget efficiency. The final actionable recommendation is to initiate operations at the North and South warehouses using the algorithmically generated routing schedule to maintain a highly resilient distribution network.

6 References

1. Mitchell, S., O'Sullivan, M., & Dunning, I. (2011). PuLP: A Linear Programming Toolkit for Python. The University of Auckland, Auckland, New Zealand.

2. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Pro-

ceedings of the 9th Python in Science Conference, 51-56.

3. City of New York. (2024). New York City Open Data: Emergency Response and Logistics Datasets [Data set]. NYC OpenData.

Abstract

This project presents a computational approach to optimizing an emergency supply distribution network for Campus City Logistics. A Mixed-Integer Linear Programming (MILP) model was developed to minimize total annual supply chain costs, which comprise fixed warehouse expenses and variable transportation costs. The model was designed to satisfy strict constraints regarding facility demand, warehouse capacity, redundancy requirements, and a $1,500,000 budget limit. Implemented in Python utilizing the PuLP library for mathematical optimization, Pandas for data processing, and Folium for geospatial mapping, the algorithm successfully identified the most cost-effective warehouse locations and delivery routes. The final results demonstrate that selecting the North and South warehouses guarantees 100 percent demand satisfaction while generating a significant financial surplus of $540,533.95, proving the efficacy of algorithmic facility location in urban logistics.